

# NBA PLAYER ARCHETYPES

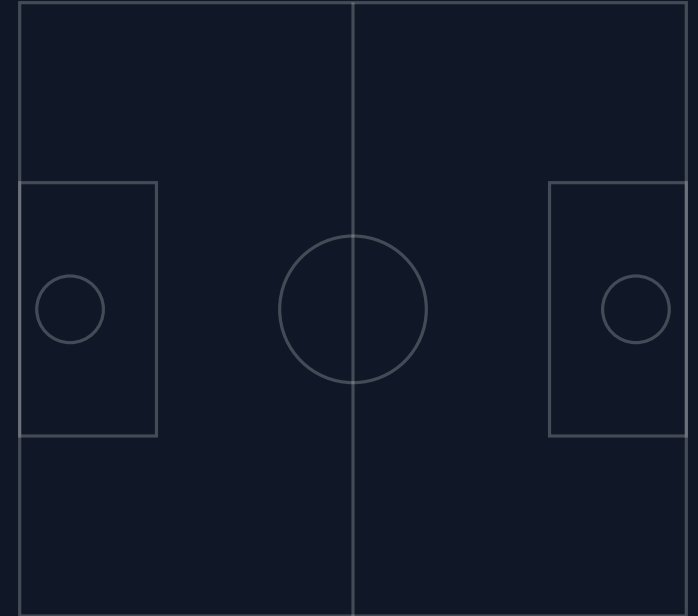
## FROM POSITIONS TO DATA-DRIVEN ROLES

A personal machine-learning project that began by questioning the modern power forward and evolved into an unsupervised clustering study of NBA player roles.

### PROJECT ARC

**Death of PF -> clustering experiments -> 10 archetypes**

2023-24 regular season | K-Means | 399-player final sample



# The project started with a position, not a cluster.

The earliest files are a "Death of PF" experiment comparing power forwards across eras. The code attempted to scrape advanced data for 1990-2005 and 2020-2023 and visualize usage, height and weight.

## ORIGINAL QUESTION

**Has the power forward changed enough that the traditional position label is becoming less useful?**

That question pushed the project toward a broader idea: instead of deciding the role first, let statistical similarity form the groups.

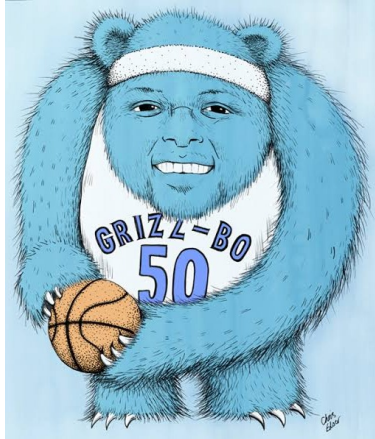


## Why the pivot mattered

Traditional positions are labels imposed before analysis. Clustering reversed the workflow: describe players with statistics, standardize the features, and ask which profiles naturally sit near one another.

# The power forward did not just get smaller. Its job changed.

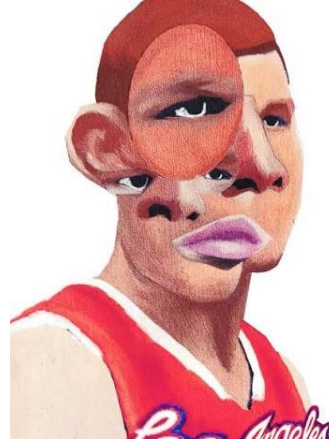
The original idea came from watching a familiar type of player disappear, then asking whether the statistics would show the same transition.



Zach Randolph - old-school power



Kenneth Faried - energy and interior play



Blake Griffin - power forward forced to adapt

## WHAT MADE THE QUESTION INTERESTING

Faried and Griffin were two favorites who had to adapt as the league changed. Randolph represents the heavier, interior-oriented PF archetype that became much harder to find.

## MY EXPECTATION

Fewer post-ups, less weight and interior power, more perimeter spacing, more shooting, and more wings occupying the nominal PF spot.

## THE BIGGER IDEA

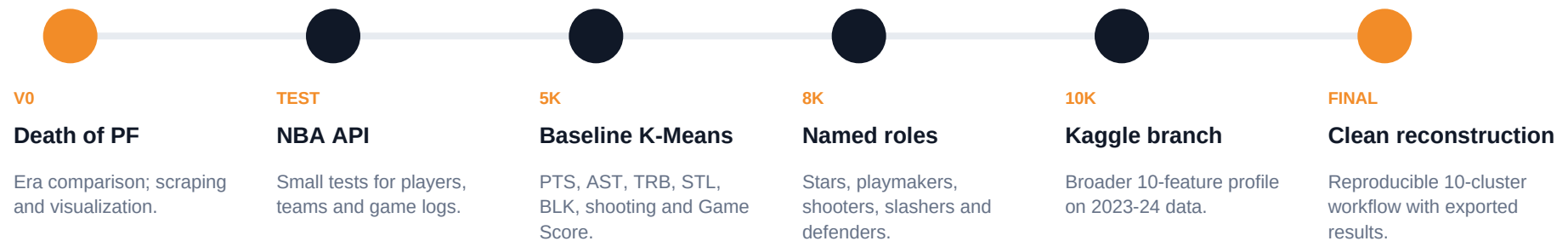
I already expected the old PF to be outdated. The more interesting question became what replaced it. Did modern power forwards converge on one new role, or did the position break apart into several roles as the NBA became more free-form?

## THE QUESTION THAT LED TO CLUSTERING

If a traditional position label no longer tells us how a player actually plays, could the data form more useful categories on its own?

# A messy path became the point of the portfolio.

The development folder preserves the original experiments instead of rewriting history. The project moved through scraping, API tests, a five-cluster baseline, an eight-role attempt, and finally a 10-cluster Kaggle branch.

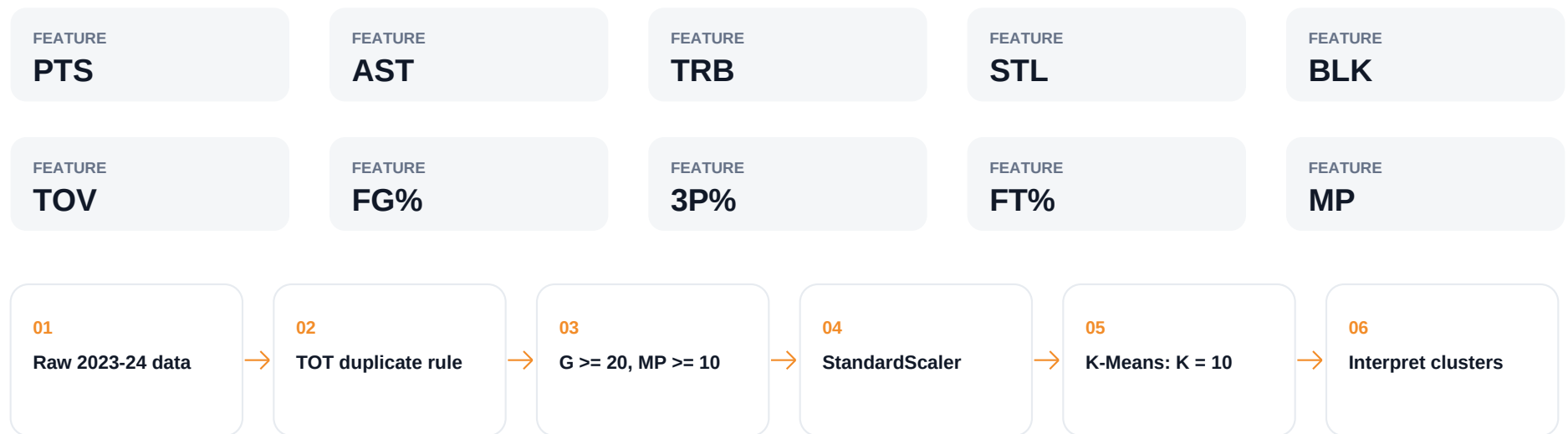


## WHAT CHANGED

The important progression was not just increasing K. The project gradually moved from a predefined basketball concept toward a data-first role taxonomy. It also exposed practical problems: choosing features, normalizing different scales, deciding how many clusters to use, handling incomplete data, and translating numeric cluster IDs into basketball language.

# The final reconstruction turns the late experiment into a reproducible pipeline.

The final script uses the same 10-feature direction found in the later development work, but adds explicit cleaning, duplicate handling, a stable random seed, 50 K-Means initializations and exported outputs.

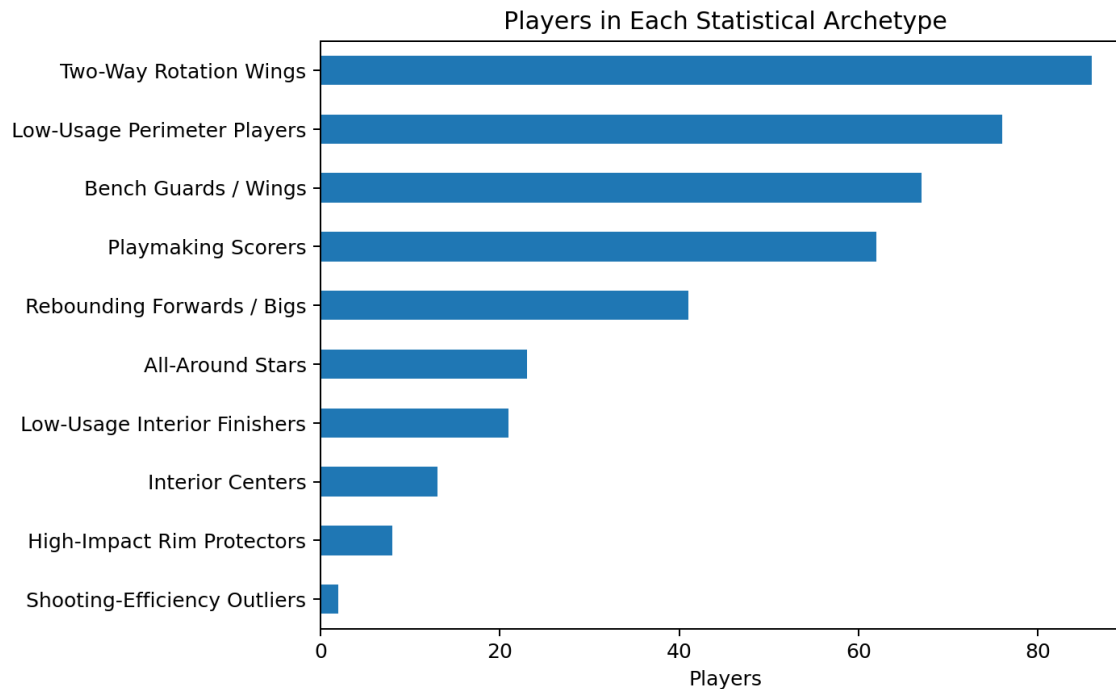


## WHY STANDARDIZE?

K-Means uses distance. Without scaling, a feature with a larger numeric range can dominate that distance. StandardScaler puts the selected statistics onto comparable standardized scales before clustering.

# Ten archetypes emerged - but they are not equally clean.

The final sample contains 399 players. Cluster names are interpretations of the statistical centroids, not labels learned by the algorithm.



## LARGEST GROUPS

Two-Way Rotation Wings 86

Low-Usage Perimeter Players 76

Bench Guards / Wings 67

Playmaking Scorers 62

Rebounding Forwards / Bigs 41

399 players after the final filters

# The better question became: what is a position actually describing?

Height, defensive matchup and lineup slot still matter, but none of them fully explains how a player contributes.

## TRADITIONAL POSITION

PG / SG / SF / PF / C gives useful lineup shorthand, but players inside the same label can have very different statistical jobs.

## STATISTICAL ROLE

Clustering asks which players resemble one another across scoring, passing, rebounding, defense, shooting efficiency and minutes - without assigning the role first.

## POSITIONLESS TREND

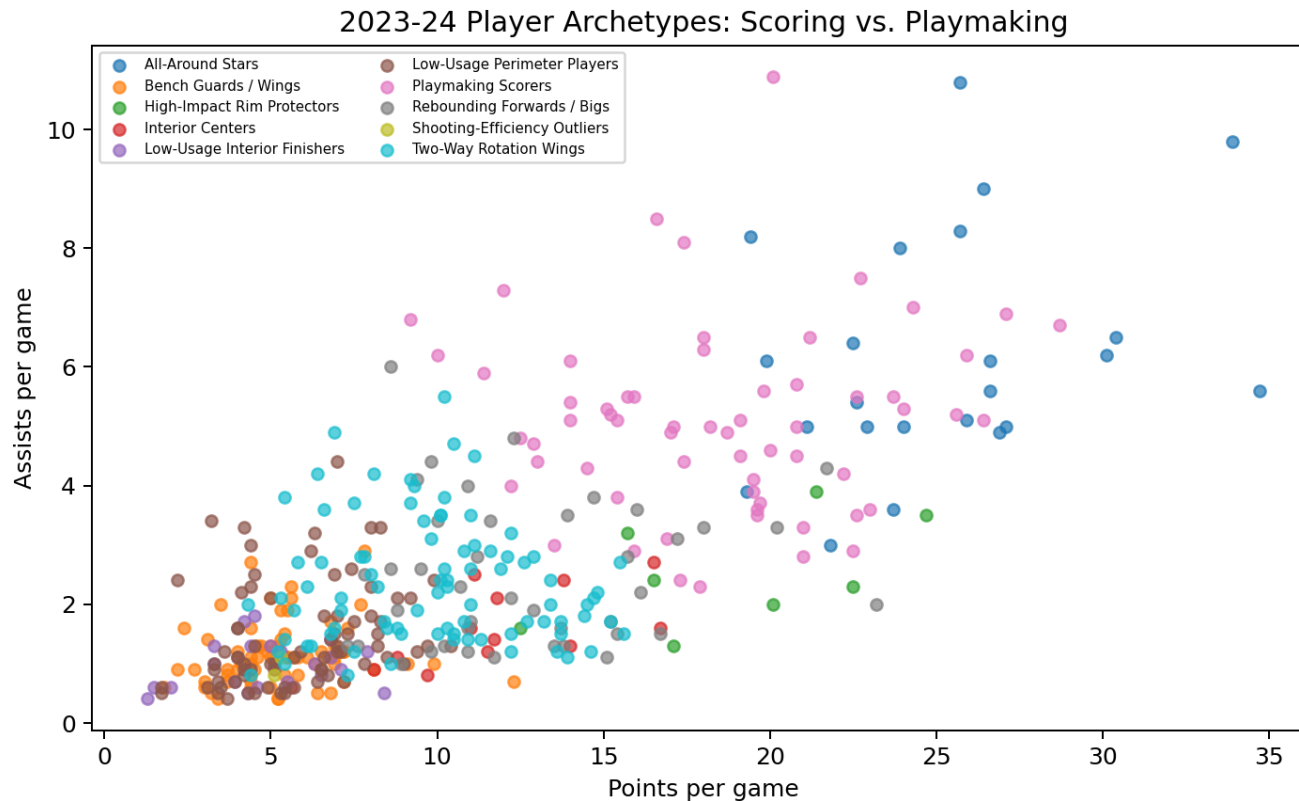
The project suggests positions still create tendencies, but they are no longer strict descriptions of style. A player can have the size of one position, guard another, and statistically resemble players listed somewhere else.



Wembanyama: size does not define a single role

# Two dimensions show separation - and why the full model needs more than two.

Plotting points versus assists gives an intuitive view of scorer/playmaker structure, but the model actually clusters in a 10-dimensional standardized feature space.



## READ THIS CAREFULLY

**Overlap on this chart does not mean the model sees the players as identical.**

Rebounding, blocks, turnovers, shooting percentages and minutes also affect cluster assignment. This plot is a projection for interpretation, not the clustering space itself.

That distinction became one of the project lessons: a clean-looking chart is not the same thing as a complete model.



# What the clusters look like in basketball language.

Representative players below are the players nearest each cluster centroid in the final output. They are examples of the statistical center of each group, not necessarily the most famous members.

| ARCHETYPE                      | N  | CENTROID-NEAR EXAMPLE | PTS  | AST | REB |
|--------------------------------|----|-----------------------|------|-----|-----|
| ↔ Two-Way Rotation Wings       | 86 | Kevin Huerter         | 10.2 | 2.6 | 3.5 |
| 📍 Low-Usage Perimeter Players  | 76 | Landry Shamet         | 7.1  | 1.2 | 1.3 |
| 🛋 Bench Guards / Wings         | 67 | Cedi Osman            | 6.8  | 1.7 | 2.5 |
| 🏀 Playmaking Scorers           | 62 | Terry Rozier          | 19.8 | 5.6 | 4.0 |
| 🏀 Rebounding Forwards / Bigs   | 41 | Wendell Carter Jr.    | 11.0 | 1.7 | 6.9 |
| ⭐ All-Around Stars             | 23 | Jayson Tatum          | 26.9 | 4.9 | 8.1 |
| ↓ Low-Usage Interior Finishers | 21 | Bruno Fernando        | 6.3  | 1.0 | 4.3 |
| 🏠 Interior Centers             | 13 | Jakob Poeltl          | 11.1 | 2.5 | 8.6 |
| 🛡 High-Impact Rim Protectors   | 8  | Chet Holmgren         | 16.5 | 2.4 | 7.9 |
| 🎯 Shooting-Efficiency Outliers | 2  | Drew Eubanks          | 5.1  | 0.8 | 4.3 |

One warning sign is the two-player "Shooting-Efficiency Outliers" cluster. A tiny cluster can be interesting, but it can also indicate that the selected features and K are isolating unusual percentage profiles rather than discovering a broadly useful basketball role.

## The clusters were not replacements for positions - they cut across them.

That distinction became one of the most useful conclusions of the project.

### ALL-AROUND STARS

Stars can come from very different listed positions. Their shared trait is statistical workload rather than one position.

### BIG MEN SPLIT APART

Centers and forwards separate into finishers, rebounders, rim protectors and higher-usage stars because their jobs differ.

### MODERN WINGS SPREAD EVERYWHERE

Players in the broad wing family can become scorers, two-way rotation pieces, low-usage perimeter players or stars. Kawhi is a useful visual example of the modern two-way wing: the label SF/PF alone does not describe the role.



## The original PF question survived - just in a different form.

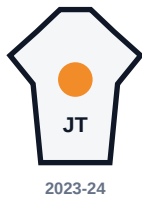
Instead of asking whether today's PF still looks like the PF of 15 or 20 years ago, the project asks where PF-listed players distribute across a role taxonomy. If they spread across many archetypes, that fragmentation is itself evidence of how flexible the position has become.

# Three statistical centers make the archetypes easier to read.

These are real 2023-24 season rows from the final exported dataset. The point is not that each player can only be one thing; it is to show what the model considered typical within three very different clusters.

## ALL-AROUND STARS

### Jayson Tatum

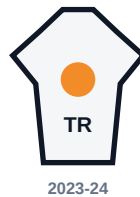


|             |            |
|-------------|------------|
| PTS         | AST        |
| <b>26.9</b> | <b>4.9</b> |
| REB         | BLK        |
| <b>8.1</b>  | <b>0.6</b> |

High scoring volume with strong minutes, rebounding and secondary creation.

## PLAYMAKING SCORERS

### Terry Rozier

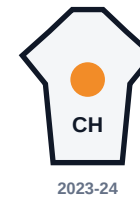


|             |            |
|-------------|------------|
| PTS         | AST        |
| <b>19.8</b> | <b>5.6</b> |
| REB         | BLK        |
| <b>4.0</b>  | <b>0.3</b> |

Guard scoring plus playmaking, with less interior impact than the star cluster.

## HIGH-IMPACT RIM PROTECTORS

### Chet Holmgren



|             |            |
|-------------|------------|
| PTS         | AST        |
| <b>16.5</b> | <b>2.4</b> |
| REB         | BLK        |
| <b>7.9</b>  | <b>2.3</b> |

Scoring is paired with an unusually strong block profile and efficient big-man shooting.

# The model discovers statistical resemblance, not basketball truth.

This became the central limitation of the project. A role is only visible to K-Means if the chosen numeric features encode it.

## WHAT THE DATA SEES WELL

- Scoring volume
- Basic playmaking
- Rebounding volume
- Steals and blocks
- Shooting percentages
- Minutes / role size

## WHAT BASIC BOX SCORES MISS

- Defensive positioning
- Off-ball movement
- Screening and spacing
- Decision quality
- Matchup versatility
- Basketball IQ / context

That means the archetypes are best read as a map of the available data. They can expose patterns, but they cannot prove that traditional positions are obsolete or fully describe a player's value.

# The final model is useful because its weaknesses are visible.

The project is exploratory. Several choices materially affect the result, and the portfolio should show those choices rather than hide them.

## K = 10

The number of clusters is chosen before fitting. Ten produced a useful level of detail, but it is not uniquely correct.

## FEATURE SELECTION

Changing the statistics changes the geometry of the player space and therefore the archetypes.

## ERA

A 2023-24 role taxonomy describes one basketball environment. The same profile may mean something different in another era.

## TINY CLUSTERS

The two-player efficiency-outlier cluster suggests some separation may reflect unusual data combinations more than a stable role.

## NAMES ARE HUMAN

K-Means outputs cluster IDs. The basketball names are interpretations assigned after inspecting the profiles.

## VALIDATION

A silhouette score can measure geometric separation, but basketball usefulness also requires domain interpretation.

# A strange cluster can teach more than a clean one.

The tiny shooting-efficiency group became a reminder that mathematical separation and basketball meaning are not the same thing.



2

PLAYERS

## WHY THIS MATTERS

A two-player cluster may reflect a genuinely unusual role, but it may also be the model reacting strongly to an extreme percentage or a small number of attempts. K-Means has no basketball intuition: it only sees distances in the feature space.

## THE MODELING LESSON

Percentages need context. A player shooting an extreme percentage on very low volume should not automatically be treated like a high-volume elite shooter. Attempt counts, shot location and richer efficiency measures would help distinguish those cases.

## THE PORTFOLIO LESSON

I would rather keep this imperfect cluster visible than hide it. It shows where the model can fail, why feature engineering matters, and why cluster names should be treated as interpretations rather than objective truths.

# The strongest outcome was the progression, not a perfect taxonomy.

The original files show a beginner learning Python and machine learning by building, testing, breaking and reframing an idea. Keeping those files makes the improvement visible.

## 01 Problem framing

A good project question can change when the data reveals a more interesting one.

## 02 Data work

Encoding, duplicates, missing values, low-minute samples and inconsistent sources matter before modeling.

## 03 Scaling

Distance-based models require attention to feature scale.

## 04 Interpretation

Cluster IDs become useful only after examining centroids and representative players.

## 05 Bias

The model reflects the statistics chosen and what basketball data is available.

## 06 Communication

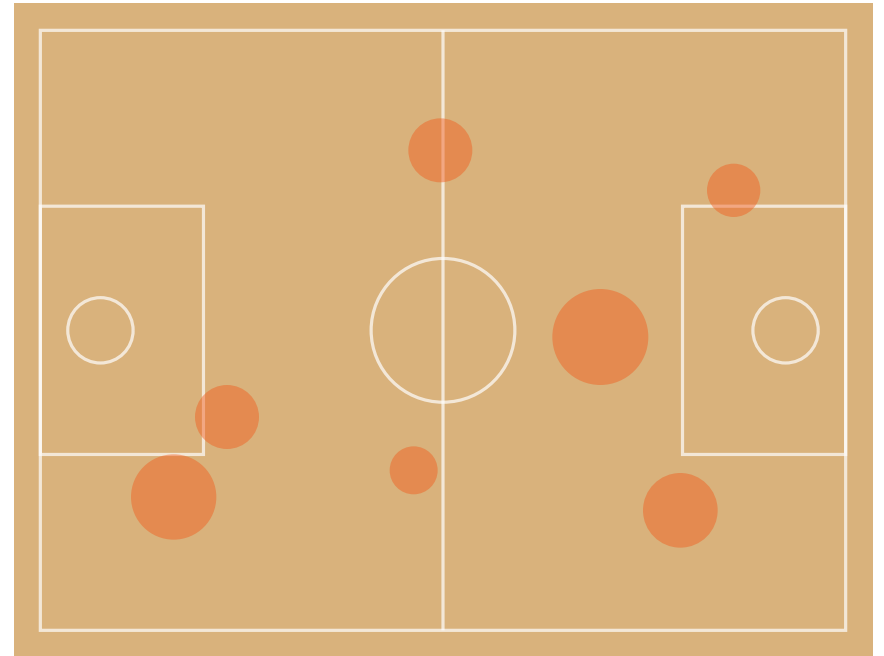
A result needs caveats, examples and visuals to become understandable outside the code.

# The next question is not just "what role is this player?"

A richer version could ask how a statistical profile changes meaning across eras and incorporate data that captures where and how actions happen, not only how often they are counted.

## FUTURE DATA

- Multiple seasons and era-specific models
- Shot-location / heat-map features
- Tracking and movement data
- Defensive matchup information
- Lineup and on/off context
- Play-type and spatial role features



The long-term idea: compare a player not only with contemporaries, but with the statistical role environment of another era.



# The next version would test whether roles survive across eras.

## MORE SEASONS



Fit comparable models across multiple seasons instead of treating 2023-24 as the whole league.

## MORE ERAS



Compare the role environment of the 1990s, 2000s, 2010s and current NBA to see which archetypes appear, disappear or change.

## MORE BASKETBALL CONTEXT



Add attempts, shot locations, play type, lineup data, defensive matchup information and tracking data so the model sees more than the box score.

## MORE ROLE DETAIL



Experiment with more categories and alternative clustering approaches, then judge them by both statistical separation and basketball usefulness.

The most interesting long-term version would return to the original power-forward idea with a better tool: instead of deciding what a PF should look like, compare how PF-listed players distribute across statistical roles in each era and watch the role map change.

# The repository separates history from presentation.

Original development files remain intact, while the final folder contains the cleaned reconstruction and exported results. This makes the portfolio reproducible without pretending the early work was cleaner than it was.

```
nba-player-archetypes/  
|-- README.md  
|-- data/  
|   |-- original project datasets  
|-- development/  
|   |-- v0_death_of_pf/  
|   |-- v1_player_clustering/  
|       |-- early_tests/  
|       |-- nba_api_tests/  
|       |-- kaggle_23_24/  
|-- final/  
|   |-- nba_player_archetypes_final.py  
|   |-- nba_2023_24_player_archetypes.csv  
|   |-- cluster_centroids_standardized.csv  
|-- portfolio/  
    |-- case study PDF
```

## PORTFOLIO PRINCIPLE

**Preserve the original work.  
Explain the progression.  
Present a clean final  
reconstruction separately.**

That distinction keeps the project honest while still making it easy for someone reviewing the repository to understand the final approach.

## FINAL TAKEAWAY

# Positions were the starting point. Statistical roles became the better question.

The project evolved from trying to describe one changing position into learning how unsupervised models can organize players - and where basic basketball data stops being enough.

## NBA PLAYER ARCHETYPES

K-Means clustering / 2023-24 NBA regular season

